Supplementary material

Asymmetric development of strategic coordination across the life cycle

Pablo Brañas-Garza[†], Diego Jorrat[‡] and Pablo Lomas[*]

A1 Experimental design and instructions

A1.1 Overview of studies

The paper draws on two studies that share the same experimental paradigm: two one-shot coordination games followed by a stated justification for each choice. The two studies share the same game design and instructions (see Section A1.2) but differ in setting and population: Study 1 is a lab-in-the-field experiment with secondary school students (ages 10–17); Study 2 consists of laboratory sessions with university-age participants (ages 17–20).

Study 1 — Lab-in-the-field, TeensLab. Study 1 constitutes a subsample of the TeensLab dataset (see Vasco et al. (2025) for details), a large-scale initiative by a consortium of Spanish universities dedicated to behavioral research with adolescents, whose data are publicly available in Vasco et al. (2025). Rather than inviting participants to a lab, we took the experiment to them. A total of 1716 students participated in this study. Between January 2023 and May 2023, we visited 20 secondary schools across two Spanish provinces — Barcelona (19 schools, 91.2% of the sample) and Cádiz (1 school, 8.8%) — and ran sessions directly inside the classrooms, with students sitting at their usual desks.

The participant pool was recruited through agreements with school headmasters, who agreed to integrate the experiment into their pedagogical curriculum and to carry it out as a classroom activity. The experiment was approved by the Ethical Committee of Universidad Loyola Andalucía (No. 20190318, 20200709 and 20230301) under an opt-out IRB consent procedure.

Participants ranged from grades 5 to 11 (ages 10–17), all enrolled in *Educación Secundaria Obligatoria* (Secondary Education in Spain) and High School. Students in vocational training programs were also recruited, reaching a total of 111 students. Additionally, 58 undergraduate students from Universidad Loyola Andalucía participated in the study as a pilot representing 3.38% of the Study 1 total sample. These university participants were aged up to 20 years old. The experiment was

[*] *LoyolaBehLab*, Universidad Loyola Andalucía.

[†] Corresponding author. Escritor Castilla Aguayo St. Nº4, 14004 Córdoba, Spain. Corresponding author. Email: branasgarza@gmail.com

[‡] Department of Applied Economics I, Universidad de Sevilla & *LoyolaBehLab*.

conducted as a regular class activity using a self-administered questionnaire programmed in a tailored online platform named SAND.[1] Subjects answered on their computers, tablets, or mobile phones with guaranteed anonymity. Throughout the main sample, payoffs were hypothetical — a deliberate choice justified by evidence that adolescents do not alter their behavior when real money is introduced (Alfonso et al., 2025).

Study 2 — Laboratory experiments. The data for Study 2 are drawn from three sessions of a laboratory experiment described in full detail in Brañas-Garza et al. (2026). All were conducted at Universidad Loyola Andalucía in designated classrooms, where participants read the instructions and made their decisions using their own mobile phones via a Qualtrics survey. Incentives were randomized at the individual level ($p = 1/2$): each participant was assigned to either a monetary or a hypothetical payment condition. Participants assigned to the hypothetical condition were not left empty-handed: the Theory of Mind task was incentivized for them, ensuring every participant had a realistic chance of earning money regardless of their assigned condition.

The first two sessions (November 2023 and November 2024, $n = 242$ in the original sample) combined two groups: 137 high school students visiting the university on an open-day event, and 105 university undergraduates enrolled at Loyola. It should be noted that the high school visitors were students who attended a university open day as a class activity of their high school. Games were presented in a fixed order (Game 1 → Game 2). Average earnings were €2.19, including a €1 show-up fee, for a session lasting under 10 minutes.

The third session (November–December 2024, $n = 301$ in the original sample) restricted recruitment to Loyola University students only and additionally randomized game order at the individual level. This double randomization — incentive scheme and game sequence — rules out the order and composition confounds that the first two sessions could not fully control. Average earnings were €1.98, again for under 10 minutes of participation. Brañas-Garza et al. (2026) show that neither game order nor the distinction between sessions affects outcomes.

For the purposes of this paper, we retain participants aged 17–20, excluding those above age 20 due to small cell sizes, as well as a small number of participants misclassified or enrolled as grade repeaters in secondary education. This yields a pooled Study 2 sample of $n = 415$.[2]

A1.2 Game design

Both games are one-shot, 2×2 coordination games with two pure-strategy Nash equilibria. Participants are told they are paired with another person in the room (someone they cannot see or communicate with) and must press one of two colored buttons simultaneously. What makes the games interesting is that the two equi-

[1] SAND (Social Analysis and Network Data; https://sand.kampal.com) is a tailored online platform designed for data collection in educational settings, enabling privacy-preserving management of large simultaneous classroom sessions.

[2] In Study 1 (TeensLab), both games were always presented in the same order: Game 1 first. This was a deliberate design choice — by exposing participants to Game 2 after Game 1, we gave RiskD its best chance, since subjects may transfer the reasoning learned in Game 1. In the third session of Study 2, game order was randomized to rule out order effects entirely; Brañas-Garza et al. (2026) confirm these have no effect on outcomes.

libria differ in the reasoning they require: one game has a clear payoff-dominant equilibrium, while the other has a risk-dominant equilibrium. The payoff values below correspond to Study 1; in Study 2 the same amounts were divided by 10, which leaves all the equilibrium structure intact.

Game 1 — Payoff-dominant equilibrium (*PayD*). Players choose between Yellow and Red. Pressing Yellow when the other person also presses Yellow gives both players 10 euros; pressing Red when the other also presses Red gives both 5 euros; and any mismatch leaves both with nothing. See payoff matrix in Table A1.

	Yellow (A)	**Red (B)**
Yellow (A)	**10, 10**	0, 0
Red (B)	0, 0	5, 5

Table A1: Game 1 — Payoff-dominant coordination game.

The logic here is: Yellow/Yellow Pareto-dominates Red/Red, and any reasonable player who expects the other to think the same way should choose Yellow.

After each game, participants were asked to choose one of four predefined reasons for their decision, presented in random order. For Game 1:

(a) Because I earned more;

(b) Because I earned less;

(c) Because I liked the colour;

(d) I chose one randomly.

We define *PayD* as the indicator for pressing Yellow *and* explaining the choice with "*Because I earned more*". This conjunction ensures we are identifying participants who are consciously playing the payoff-dominant equilibrium, not just pressing a button at random.

Game 2 — Risk-dominant equilibrium (*RiskD*). Players choose between White and Blue. Both White/White and Blue/Blue pay $(10, 10)$, so payoffs cannot guide the decision. The difference shows up only in miscoordination: pressing White when the other presses Blue gives you 5 euros and them nothing; pressing Blue when the other presses White gives you nothing and them 5 euros. White is therefore the risk-dominant choice. See payoff matrix of Game 2 in Table A2.

	White (A)	**Blue (B)**
White (A)	**10, 10**	5, 0
Blue (B)	0, 5	10, 10

Table A2: Game 2 — Risk-dominant coordination game.

For Game 2, the four predefined reasons were:

(a) Because it was less risky;

(b) *Because it was more risky;*

(c) *Because I liked the colour;*

(d) *I chose one randomly.*

We define *RiskD* as pressing White *and* choosing the reasoning "*Because it was less risky*". Again, the conjunction is essential: it filters out participants who happen to press White for aesthetic reasons or by pure chance.

The full experimental sequence was always: Game 1 → Explanation 1 → Game 2 → Explanation 2 (with game order randomized only in session 3 of Study 2).

A1.3 Instructions

Below we reproduce the exact text shown to participants on each of the six screens. The wording is identical across both studies; the amounts shown in the instructions were divided by 10 for Study 2, leaving all the equilibrium structure intact. In Study 1, the screens appeared in Spanish; the following is a faithful English translation. Button colors were not randomized: the equilibrium options were always Yellow for Game 1 and White for Game 2, ensuring the salience structure was consistent across participants. Button positions — left or right — were randomized independently for each participant to prevent any systematic positional bias.

Game 1 screens

Screen 1 (Introduction):

You will now be paired with a random partner.

Both of you will need to press a button without being able to communicate with each other.

Depending on what each of you chooses, you will receive different (hypothetical) rewards.

[Click the forward button to move on to the first question.]

Screen 2 (Game 1 — payoff matrix):

These are the possible outcomes:

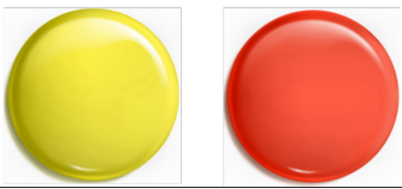
If you press yellow and she/he presses yellow, you both win 10 euros each.

If you press yellow and she/he presses red, you both win 0 euros each.

If you press red and she/he presses yellow, you both win 0 euros each.

If you press red and she/he presses red, you both win 5 euros each.

Which button do you press?



Screen 3 (Explanation 1):

Why did you choose the previous button?

○ Because I earned more
○ Because I earned less
○ Because I liked the colour
○ I chose one randomly

Game 2 screens

Screen 4 (Transition):

Now you are going to play again with another person, other buttons, and other prizes.

Again, you have to choose a button without knowing what the other one does.

Screen 5 (Game 2 — payoff matrix):

These are the possible outcomes:

If you press white and she/he presses white, you both win 10 euros each.

If you press white and she/he presses blue, you will earn 5 euros and she/he will win 0 euros.

If you press blue and she/he presses blue, you both win 10 euros each.

If you press blue and she/he presses white, you will earn 0 euros and she/he will win 5 euros.

Which button do you press?

Screen 6 (Explanation 2):

Why did you choose the previous button?

○ Because it was less risky
○ Because it was more risky
○ Because I liked the colour
○ I chose one randomly

A1.4 Tasks

Beyond the two coordination games, participants completed a short battery of tasks designed to capture cognitive abilities and, in the developmental sample, individual risk preferences. The battery was not identical across studies — some tasks require a minimum level of abstraction that makes them unsuitable for 10-year-olds, and others are redundant once we have university students. Table A3 summarizes which tasks appeared where.

Task	Study 1 (TeensLab)	Study 2 (Qualtrics)
Cognitive Reflection Test (CRT)	✓	✓
Financial Abilities (FinAb)	✓	—
Probabilistic Reasoning (Del)	✓	—

Table A3: Tasks administered in each study.

A1.4.1 Cognitive Reflection Test (CRT)

The CRT was administered in both studies. We used the adaptation by Thomson and Oppenheimer (2016), as in Alfonso et al. (2025), which preserves the spirit of Frederick (2005)'s original three-item test while reducing reliance on domain-specific vocabulary that could disadvantage younger or less math-oriented respondents. Each item has a tempting intuitive answer that is wrong, and the correct answer requires a moment of deliberate reflection. Items were randomized in order. The score runs from 0 (all intuitive) to 3 (all reflective). The three items are reproduced below.

Q1. Emilia's father has three daughters. The first two are named April and May. What is the name of the third daughter?

(Reflective: **Emilia**; Intuitive: June)

Q2. If you are running a race and you overtake the person in second place, what position are you in? Indicate with a number (e.g., 1 = first, 2 = second, etc.).

(Reflective: **2**; Intuitive: 1)

Q3. In a library, the number of books doubles every month. If it takes 48 months for the library to become full, how many months would it take for it to be half full? Indicate with a number.

(Reflective: **47**; Intuitive: 24)

A1.4.2 Financial Abilities (FinAb)

In Study 1, we complemented the CRT with a short financial literacy task.[3] The three items require basic arithmetic with money and an understanding of how interest compounds over time. Scores run from 0 to 3.

Q1. You have 5 euros and you spend 3 euros. How many euros do you have left?

(Correct answer: **2 euros**)

Q2. You put 100 euros in a savings account. The bank gives you 10% interest per year. How much money will you have after one year?

(Correct answer: **110 euros**)

[3]Items are drawn from Alfonso et al. (2025). They span basic arithmetic with money and simple compound interest, and were calibrated so as not to require knowledge beyond the primary school curriculum.

Q3. You put 100 euros in a savings account that pays 10% interest per year. You leave the money there for two years without withdrawing anything. How much will you have at the end of the second year?

(Correct answer: **121 euros**)

A1.4.3 Probabilistic Reasoning (Del)

Three probabilistic reasoning items (*Del*) adapted from Delavande and Rohwedder (2008) were administered exclusively in Study 1. Participants received one point for each item in which they correctly respected a basic probabilistic relation, for a total score ranging from 0 to 3. The three scored comparisons were: (i) stating that the probability of drawing the green apple from a basket of 5 (1 green, 4 red) is higher than from a basket of 10 (1 green, 9 red); (ii) stating that the probability of eating rice in the next month is higher than in the next week; and (iii) stating that the probability of showering at least once in the next month is higher than the probability of not attending school for a single day in the same period. Violations occur when the reported probability for the broader or more likely event is numerically lower than that for the narrower or less likely event.

Participants responded by entering a number from 0 to 100 (50% of participants) or by moving a slider on a 0–100 scale (50% of participants); randomization was at the individual level. The instructions and questions were identical across both response formats. Questions (Q1–Q6) are shown below.

Screen 1 — Instructions

In the next screen, we **will ask you** about the **probability** that certain things could happen. In each question, you should **write down** how **likely you think** it is that these things will happen in reality.

If you think something will happen for **sure**, you should write **100**. If you are fairly but not completely sure, write a number close to 100 (but less than 100). If you think it is something that will **never** happen, write **0**. If you think it is unlikely, write a number close to 0 (but greater than 0). If you think it is **equally likely** to happen or not to happen, write **50**.

Screen 2 — Examples

EXAMPLES:

• If it is a sunny day and I ask you how likely it is that it will be sunny within 1 hour, you would typically say 100 or numbers very close to 100.

• If it is 12 o'clock at night, the most reasonable thing to say is that the probability that it will be sunny in 1 hour is 0.

• If I flip a coin, the probability that it will come up heads is 50.

***Del* questions (2 by screen)**

Q1. Imagine that I have a basket with 5 apples: 1 green and 4 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q2. Imagine that I have a basket with 10 apples: 1 green and 9 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q3. How likely is it that you eat rice in the next week from 0 to 100? (Including today)

Q4. How likely is it that you eat rice in the next month from 0 to 100? (Including today)

Q5. How likely is it that you do not attend your educational center even once in the next month from 0 to 100? (Including today)

Q6. How likely is it that you take a shower at least once during the next month from 0 to 100? (Including today)

The slider version (used by 50% of participants, randomized at the individual level) presented the same questions and instructions, with participants dragging a marker along a 0–100 scale ("Valor: 50" shown by default) instead of typing a number. See Figure A1 for an illustration of the slider interface.

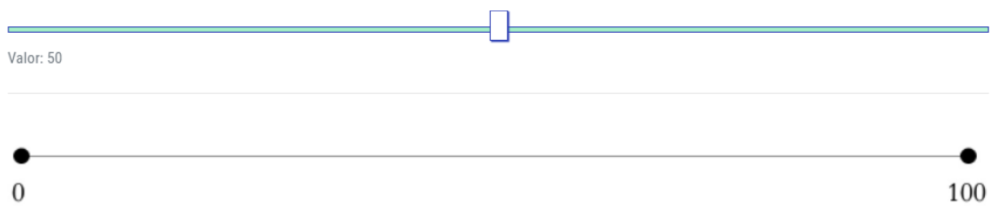


Figure A1: Slider interface. *Screenshot of the slider response format used by half of the participants in the probabilistic reasoning task. Participants dragged a marker along a 0–100 scale; the default position was 50 ("Valor: 50"). The other half of participants typed a number directly.*

A2 Sample composition

This appendix provides a full breakdown of participants across both studies. The samples span a wide range of educational stages, from late primary school to university, generating substantial variation in age, cognitive development, and educational experience. Each study's sample is described along two complementary dimensions. First, we report age in years (Tables A4 and A6), which provides a continuous developmental scale that facilitates comparison across cohorts and studies. Second, we report participants' academic track or grade (Tables A5 and A7), which reflects the structure of the educational system used for recruitment. In the Spanish system, grades 5–6 correspond to late primary education, grades 7–10 to compulsory secondary education, and grades 11–12 to academic high school. Vocational training (VT) and university (U) students are reported as separate categories, as they represent distinct educational pathways relative to the standard academic track. Table A8 presents the distribution of our full sample by age in years.

	10	11	12	13	14	15	16	17	18	19	20	Total
N	136	208	237	276	300	244	124	24	78	53	36	**1,716**
% of sample	7.9%	12.1%	13.8%	16.1%	17.5%	14.2%	7.2%	1.4%	4.6%	3.1%	2.1%	**100.0%**
% Female	53.7%	51.4%	43.5%	46.0%	47.0%	52.5%	43.5%	50.0%	60.3%	43.4%	55.6%	**48.7%**

Table A4: Study 1 — Sample distribution by age.

	5th	6th	7th	8th	9th	10th	11th	12th	VT	U	Total
N	173	213	259	299	340	190	73	–	111	58	**1,716**
% of sample	10.1%	12.4%	15.1%	17.4%	19.8%	11.1%	4.2%	–	6.5%	3.4%	**100.0%**
% Female	53.8%	47.4%	47.5%	45.8%	45.6%	53.2%	46.6%	–	70.3%	22.4%	**48.7%**

Table A5: Study 1 — Sample distribution by grade.

	17	18	19	20	Total
N	119	79	144	73	**415**
% of sample	28.7%	19.0%	34.7%	17.6%	**100.0%**
% Female	54.6%	40.5%	47.2%	46.3%	**46.3%**

Table A6: Study 2 — Sample distribution by age.

	12th	University	Total
N	119	296	**415**
% of sample	28.7%	71.3%	**100.0%**
% Female	54.6%	42.9%	**46.3%**

Table A7: Study 2 — Sample distribution by grade.

	10	11	12	13	14	15	16	17	18	19	20	Total
N	136	208	237	276	300	244	124	143	157	197	109	**2,131**
% of sample	6.4%	9.8%	11.1%	12.9%	14.1%	11.4%	5.8%	6.7%	7.4%	9.2%	5.1%	**100.0%**
% Female	53.7%	51.4%	43.5%	46.0%	47.0%	52.5%	43.5%	53.8%	50.3%	46.2%	43.1%	**48.3%**

Table A8: Full sample — All studies combined, by age.

A3 Additional data analysis

A3.1 Non-parametric analysis of risk-based coordination

Figure A2 presents non-parametric lowess (Locally Weighted Scatterplot Smoothing) estimates of the relationship between RiskD and two key predictors. Panel A shows smoothed RiskD trajectories by age separately for high- and low-ability individuals (bandwidth = 0.8). The developmental gradient for high-ability individuals is steeper and emerges earlier, rising continuously from age 10, while low-ability individuals display a more gradual increase that accelerates only in the later stages of the developmental window. Panel B shows the smoothed relationship between RiskD and age-residualized cognitive ability CogAb(r) across the full sample (bandwidth = 0.6). The relationship is markedly non-linear: RiskD remains largely flat below the median (CogAb(r) = 0, dashed vertical line) and rises sharply above it, motivating the inclusion of the quadratic term CogAb(r)^2 in columns 4 and 8 of Table 1.

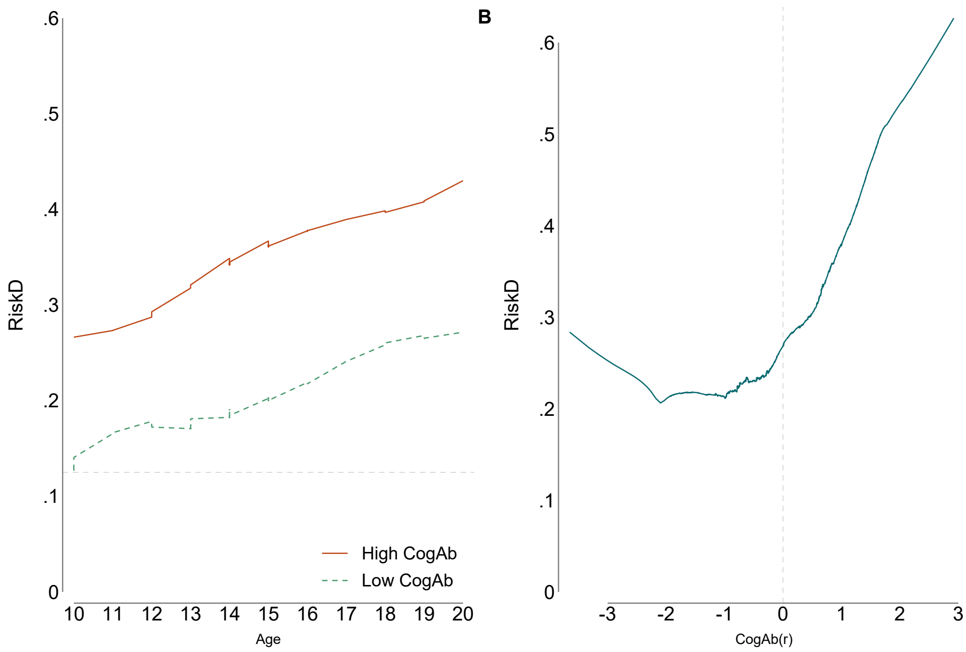


Figure A2: Lowess-smoothed relationships between deliberate risk-based coordination, age, and cognitive ability.

We further explore the relationship between cognitive ability and risk-dominant reasoning. Figure A2 shows that RiskD increases monotonically with age and is consistently higher among individuals with greater cognitive ability. This pattern is not driven by arbitrary group definitions: exploiting the full continuous variation in cognitive ability reveals a non-linear relationship, with a marked increase in RiskD above the sample mean.

Taken together, these results suggest that risk-dominant coordination is cognitively demanding and becomes more prevalent as individuals develop both in age and cognitive ability. Importantly, this pattern complements the age gradients documented above, indicating that developmental changes in RiskD are not solely driven by age per se but are closely linked to underlying cognitive skills.

A3.2 Robustness checks and additional regression analysis

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.009	0.009	0.007	0.009	0.015	0.015	0.015	0.015
	(0.011)	(0.011)	(0.013)	(0.011)	(0.010)	(0.010)	(0.012)	(0.010)
CogAb(r)	0.093***	−0.078	0.093***	0.092***	0.087***	0.014	0.087***	0.087***
	(0.014)	(0.119)	(0.014)	(0.014)	(0.013)	(0.103)	(0.013)	(0.013)
Female	−0.035	−0.034	−0.092	−0.037	−0.021	−0.020	−0.008	−0.020
	(0.025)	(0.025)	(0.192)	(0.025)	(0.023)	(0.023)	(0.173)	(0.023)
Age × CogAb(r)		0.013				0.006		
		(0.009)				(0.008)		
Age × Female			0.004				−0.001	
			(0.015)				(0.013)	
CogAb(r)^2				−0.014				0.005
				(0.013)				(0.011)
Constant	0.425*	0.411*	0.451*	0.427*	0.179	0.174	0.174	0.179
	(0.227)	(0.227)	(0.244)	(0.226)	(0.208)	(0.209)	(0.221)	(0.208)
Observations	1,474	1,474	1,474	1,474	1,462	1,462	1,462	1,462
R-squared	0.083	0.085	0.083	0.084	0.062	0.063	0.062	0.063
BIC	2108.8	2113.9	2116.0	2114.9	1812.6	1819.4	1819.9	1819.7
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table A9: OLS regression for the sub-sample: Late primary + Secondary. *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.015	0.015	0.012	0.015	0.021**	0.021**	0.029***	0.021**
	(0.010)	(0.010)	(0.012)	(0.010)	(0.009)	(0.009)	(0.011)	(0.009)
CogAb(r)	0.085***	0.073	0.085***	0.084***	0.080***	0.122	0.080***	0.081***
	(0.013)	(0.095)	(0.013)	(0.013)	(0.012)	(0.085)	(0.012)	(0.012)
Female	−0.032	−0.031	−0.117	−0.033	−0.040*	−0.040*	0.168	−0.038*
	(0.023)	(0.024)	(0.161)	(0.024)	(0.022)	(0.022)	(0.147)	(0.022)
Age × CogAb(r)		0.001				−0.003		
		(0.007)				(0.006)		
Age × Female			0.006				−0.015	
			(0.012)				(0.011)	
CogAb(r)^2				−0.009				0.011
				(0.011)				(0.010)
Constant	0.588***	0.587***	0.629***	0.594***	0.114	0.118	0.013	0.106
	(0.161)	(0.161)	(0.179)	(0.161)	(0.149)	(0.150)	(0.165)	(0.149)
Observations	1,666	1,666	1,666	1,666	1,653	1,653	1,653	1,653
R-squared	0.076	0.076	0.076	0.077	0.060	0.060	0.061	0.061
BIC	2397.8	2405.2	2404.9	2404.5	2097.3	2104.4	2102.6	2103.4
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table A10: OLS regression for the sub-sample: Late primary + Secondary + High School. *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.016	0.016	0.016	0.016	0.021**	0.021**	0.024**	0.021**
	(0.010)	(0.010)	(0.011)	(0.010)	(0.009)	(0.009)	(0.010)	(0.009)
CogAb(r)	0.084***	0.087	0.084***	0.083***	0.082***	0.100	0.081***	0.083***
	(0.013)	(0.082)	(0.013)	(0.013)	(0.012)	(0.073)	(0.012)	(0.012)
Female	−0.036	−0.036	−0.043	−0.038*	−0.034	−0.034	0.042	−0.032
	(0.023)	(0.023)	(0.140)	(0.023)	(0.021)	(0.021)	(0.124)	(0.021)
Age × CogAb(r)		−0.000				−0.001		
		(0.006)				(0.005)		
Age × Female			0.001				−0.006	
			(0.010)				(0.009)	
CogAb(r)^2				−0.013				0.014
				(0.011)				(0.010)
Constant	0.572***	0.572***	0.575***	0.581***	0.108	0.109	0.070	0.097
	(0.159)	(0.159)	(0.173)	(0.159)	(0.147)	(0.148)	(0.159)	(0.148)
Observations	1,777	1,777	1,777	1,777	1,764	1,764	1,764	1,764
R-squared	0.084	0.084	0.084	0.085	0.061	0.061	0.062	0.063
BIC	2561.2	2568.7	2568.7	2567.3	2233.1	2240.5	2240.2	2238.3
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table A11: OLS regression for the sub-sample: Late primary + Secondary + High School + Vocational. *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) PayD	(2) PayD	(3) RiskD	(4) RiskD
Age	0.021**	0.021**	0.030***	0.029***
	(0.010)	(0.010)	(0.009)	(0.009)
CogAb(r)	0.163***	0.157***	0.135***	0.106**
	(0.054)	(0.056)	(0.050)	(0.051)
Female	0.030	0.028	0.127	0.120
	(0.108)	(0.108)	(0.099)	(0.099)
CogAb(r)^2		0.003		0.017**
		(0.008)		(0.008)
Age × CogAb(r)	−0.006*	−0.005	−0.005	−0.002
	(0.003)	(0.004)	(0.003)	(0.003)
Age × Female	−0.005	−0.005	−0.012*	−0.012*
	(0.007)	(0.007)	(0.007)	(0.007)
Age × Female × CogAb(r)	−0.000	−0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Constant	0.547***	0.545***	0.046	0.036
	(0.158)	(0.158)	(0.147)	(0.147)
Observations	2,131	2,131	2,118	2,118
R-squared	0.080	0.080	0.068	0.070
BIC	3053.4	3060.9	2756.0	2758.5
School FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Table A12: OLS regressions of deliberate coordination on age, cognitive ability, and gender. *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.019**	0.018*	0.021**	0.018**	0.025***	0.025***	0.029***	0.025***
	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)
CogAb(r)	0.067***	0.156***	0.067***	0.071***	0.069***	0.170***	0.068***	0.072***
	(0.010)	(0.055)	(0.010)	(0.011)	(0.010)	(0.052)	(0.010)	(0.009)
Female	−0.048**	−0.049**	0.038	−0.046**	−0.051***	−0.052***	0.108	−0.047**
	(0.021)	(0.021)	(0.108)	(0.021)	(0.019)	(0.019)	(0.102)	(0.019)
Age × CogAb(r)		−0.006*				−0.006*		
		(0.003)				(0.003)		
Age × Female			−0.006				−0.011	
			(0.007)				(0.007)	
CogAb(r)^2				0.009				0.016**
				(0.008)				(0.007)
Observations	2,131	2,131	2,131	2,131	2,118	2,118	2,118	2,118
CogAbB2	No	No	No	Yes	No	No	No	Yes
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table A13: Logit regressions of deliberate coordination on age, cognitive ability, and gender. *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A14 replicates Table 1 using MICE-PMM imputation ($M = 20$). All main findings are qualitatively unchanged. The age gradient in RiskD remains positive and significant across all specifications, and cognitive ability remains a strong positive predictor of both outcomes. In column 6, the coefficient on CogAb(r) is 0.092 ($p = 0.106$), compared to 0.138 ($p < 0.01$) in Table 1 — the difference reflects the additional uncertainty introduced by imputation for university participants rather than a change in the underlying effect. All interaction terms retain their sign and direction.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.019**	0.019**	0.022**	0.019**	0.024***	0.024***	0.031***	0.024***
	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
CogAb(r)	0.073***	0.101*	0.072***	0.072***	0.073***	0.092	0.072***	0.074***
	(0.011)	(0.061)	(0.011)	(0.011)	(0.010)	(0.057)	(0.010)	(0.010)
Female	−0.045**	−0.046**	0.035	−0.046**	−0.052***	−0.052***	0.135	−0.050**
	(0.021)	(0.021)	(0.108)	(0.021)	(0.020)	(0.020)	(0.101)	(0.020)
Age × CogAb(r)		−0.002				−0.001		
		(0.004)				(0.004)		
Age × Female			−0.006				−0.013*	
			(0.007)				(0.007)	
CogAb(r)^2				−0.005				0.011
				(0.009)				(0.008)
Constant	0.573***	0.575***	0.536***	0.579***	0.121	0.122	0.033	0.109
	(0.147)	(0.147)	(0.155)	(0.148)	(0.139)	(0.139)	(0.146)	(0.139)
Observations	2,131	2,131	2,131	2,131	2,118	2,118	2,118	2,118
Imputation	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM
M	20	20	20	20	20	20	20	20
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table A14: Replication of Table 1 using multiple imputation by chained equations (MICE-PMM, $M = 20$ datasets). *All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

References

Alfonso, A., Brañas-Garza, P., Jorrat, D., Prissé, B., & Vázquez-De Francisco, M. J. (2025). The baking of preferences throughout the high school. *Journal of Behavioral and Experimental Economics*, 102396.

Brañas-Garza, P., Jorrat, D., & Lomas, P. (2026). *Coordination with and without incentives* [Manuscript under review].

Delavande, A., & Rohwedder, S. (2008). Eliciting subjective probabilities in internet surveys. *Public Opinion Quarterly*, *72*(5), 866–891.

Frederick, S. (2005). Cognitive reflection and decision making. *Journal of Economic Perspectives*, *19*(4), 25–42.

Thomson, K. S., & Oppenheimer, D. M. (2016). Investigating an alternate form of the cognitive reflection test. *Judgment and Decision Making*, *11*(1), 99–113.

Vasco, M., Alfonso, A., Arenas, A., Cabrales, A., Cuesta, J. A., Espín, A. M., Espinosa, M. P., García-Muñoz, T., Izquierdo Nofuentes, À., Jorrat, D., et al. (2025).

Economic preferences and cognitive abilities among teenagers in spain. *Scientific Data*, *12*(1), 7.